

Empirical Methods in Finance

Section 12 - Change in Regimes and Structural Breaks

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Objectives of the class

At the end of the class, students should be able to:

- distinguish structural breaks from returning regimes;
- test for one known break, one unknown break and multiple breaks;
- understand the logic of the Bai-Perron methodology;
- write and estimate a simple Markov-switching model;
- explain why a change in the correlation regime changes the Markowitz portfolio.

Core idea: in finance, parameters do not only fluctuate - they can also jump or switch between states.

1. Structural breaks: theory, tests and interpretation
2. Markov-switching models: latent states and filtering
3. Existing applications: business cycles and market regimes
4. New application: correlation regimes and the optimal Markowitz portfolio

Structural breaks

Why structural breaks matter

In a stable linear model, the same coefficients describe the entire sample:

$$y_t = x_t' \beta + u_t, \quad t = 1, \dots, T. \quad (1)$$

A structural break means that this parameter vector changes at one or several dates:

$$y_t = x_t' \beta_j + u_t, \quad T_{j-1} < t \leq T_j. \quad (2)$$

- The model can look correct locally and wrong globally.
- Forecasting models may fail when estimated across incompatible regimes.
- Portfolio decisions are directly affected because means, volatilities and correlations enter optimization.

Breaks versus regimes

Structural break	Regime switching
One DGP is replaced by another	The DGP can return later
Break dates are usually few	States can alternate many times
Break date may be known or unknown	State is usually latent
Bai-Perron, Chow, sup-Wald	Hamilton filter, Markov chain
Example: post-2020 equity-bond correlation	Example: bull/bear market probabilities

Practical distinction: a break is about permanent parameter instability; a regime is about recurrent parameter instability.

Known break date: the Chow test

Assume a candidate break date T_b is known. Estimate one pooled regression and two sub-sample regressions:

$$y_t = x_t' \beta + u_t, \quad t = 1, \dots, T, \quad (3)$$

$$y_t = x_t' \beta_1 + u_t, \quad t = 1, \dots, T_b, \quad (4)$$

$$y_t = x_t' \beta_2 + u_t, \quad t = T_b + 1, \dots, T. \quad (5)$$

The null hypothesis is parameter stability:

$$H_0 : \beta_1 = \beta_2. \quad (6)$$

Reference: Chow (1960), *Econometrica*.

The Chow statistic

Let SSR_0 be the residual sum of squares of the pooled regression and SSR_1, SSR_2 the residual sums of squares of the two sub-sample regressions. With k regressors:

$$F_{Chow} = \frac{(SSR_0 - SSR_1 - SSR_2)/k}{(SSR_1 + SSR_2)/(T - 2k)}. \quad (7)$$

Under the null and classical assumptions:

$$F_{Chow} \sim F(k, T - 2k). \quad (8)$$

- Intuition: do we lose too much fit by imposing one common parameter vector?
- Limitation: the break date must be known before running the test.

Unknown break date: Quandt-Andrews logic

When the break date is unknown, repeat the Chow test over admissible candidate dates:

$$\Lambda_T = \{T_0, T_0 + 1, \dots, T - T_0\}, \quad (9)$$

where T_0 is a trimming parameter used to avoid very short sub-samples. The Quandt likelihood ratio / sup-Wald statistic is:

$$\sup F = \sup_{\tau \in \Lambda_T} F_T(\tau). \quad (10)$$

- This solves the known-date problem mechanically.
- The critical values are non-standard because τ is unidentified under H_0 .

References: Quandt (1960), Andrews (1993), Andrews and Ploberger (1994).

Wald form of the unknown-break test

For a candidate break fraction $\lambda = T_b/T$, write the restricted null as:

$$H_0 : R(\lambda)\theta = 0, \quad (11)$$

where θ stacks the pre-break and post-break coefficients. The Wald statistic is:

$$W_T(\lambda) = T \left[R(\lambda)\hat{\theta} \right]' \left[R(\lambda)\hat{V}(\hat{\theta})R(\lambda)' \right]^{-1} \left[R(\lambda)\hat{\theta} \right]. \quad (12)$$

The unknown-date test is then:

$$\sup W_T = \sup_{\lambda \in [\varepsilon, 1-\varepsilon]} W_T(\lambda). \quad (13)$$

CUSUM tests: gradual instability

Another family of tests relies on recursive residuals. For $t > k$, define:

$$w_t = \frac{y_t - x_t' \hat{\beta}_{t-1}}{\sqrt{1 + x_t'(X_{t-1}' X_{t-1})^{-1} x_t}}. \quad (14)$$

The cumulative sum process is:

$$C_t = \frac{1}{\sqrt{t}} \sum_{i=k+1}^t w_i. \quad (15)$$

- If the model is stable, C_t behaves like a Brownian bridge.
- If C_t crosses confidence bands, the model is unstable.
- Useful as a diagnostic, less precise for estimating multiple dates.

Reference: Brown, Durbin and Evans (1975), *JRSS B*.

Multiple breaks: the Bai-Perron model

Bai and Perron allow for m unknown breaks in a linear regression:

$$y_t = x_t' \beta + z_t' \delta_j + u_t, \quad T_{j-1} < t \leq T_j, \quad (16)$$

with:

$$j = 1, \dots, m+1, \quad T_0 = 0, \quad T_{m+1} = T. \quad (17)$$

- $x_t' \beta$ contains stable regressors.
- $z_t' \delta_j$ contains regressors whose coefficients can break.
- A pure mean-break model is obtained when $z_t = 1$.

References: Bai and Perron (1998, 2003).

Break-date estimation as a global minimization

For a fixed number of breaks m , the break dates are estimated by minimizing the total sum of squared residuals:

$$(\hat{T}_1, \dots, \hat{T}_m) = \arg \min_{(T_1, \dots, T_m)} \sum_{j=1}^{m+1} SSR(T_{j-1} + 1, T_j). \quad (18)$$

For each segment:

$$SSR(a, b) = \min_{\delta_j} \sum_{t=a}^b \left(y_t - x'_t \hat{\beta} - z'_t \delta_j \right)^2. \quad (19)$$

- The minimization is global, not greedy.
- Dynamic programming makes the search computationally feasible.

Why dynamic programming helps

Let $C(r, T)$ be the minimum SSR for r segments ending at date T . The recursion is:

$$C(r, T) = \min_{h \leq s \leq T-h} \{C(r-1, s) + SSR(s+1, T)\}, \quad (20)$$

where h is the minimum segment length.

- Without dynamic programming, the number of partitions explodes.
- With dynamic programming, the optimal partition can be recovered backwards.
- This is why Bai-Perron is practical even with several breaks.

Testing the number of breaks

Bai and Perron propose several tests. The most useful intuition is sequential:

$$H_0 : m = \ell \quad \text{against} \quad H_1 : m = \ell + 1. \quad (21)$$

The statistic compares the fit with ℓ and $\ell + 1$ breaks:

$$F_T(\ell + 1|\ell) = \frac{SSR_\ell - SSR_{\ell+1}}{\hat{\sigma}^2} \times \text{normalization}. \quad (22)$$

Two global tests are often reported:

$$UD \max = \max_{1 \leq m \leq M} F_T(m), \quad (23)$$

$$WD \max = \max_{1 \leq m \leq M} w_m F_T(m). \quad (24)$$

- *UD* max tests no break against an unknown number of breaks.
- *WD* max reweights tests to improve power across alternatives.

Confidence intervals for break dates

Break dates are estimated with uncertainty. Under standard assumptions, the estimated break fraction converges faster than ordinary parameters:

$$\hat{\lambda}_j = \frac{\hat{T}_j}{T}, \quad \hat{\lambda}_j - \lambda_j = O_p(T^{-1}). \quad (25)$$

- Coefficients converge at the usual $\sqrt{T}$ rate.
- Break dates converge at rate T when breaks are large enough.
- Small breaks are harder: confidence intervals can become wide.

Interpretation: a break date is an econometric estimate, not a historical label. The economics still matters.

Breaks in second moments

Many financial applications are not about the conditional mean, but about variance and covariance. For a vector of returns r_t :

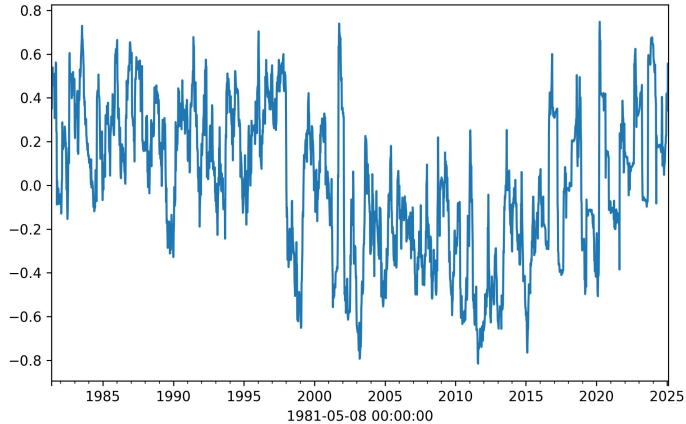
$$\Sigma_j = \mathbb{E}[(r_t - \mu_j)(r_t - \mu_j)'], \quad T_{j-1} < t \leq T_j. \quad (26)$$

For two assets i and k , the correlation in segment j is:

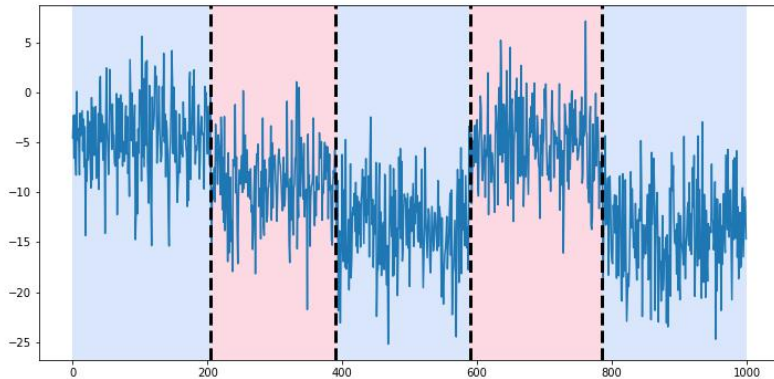
$$\rho_{ik,j} = \frac{\sigma_{ik,j}}{\sqrt{\sigma_{ii,j}\sigma_{kk,j}}}. \quad (27)$$

- A break in correlation can transform diversification properties.
- This is often more important for portfolio construction than a break in mean returns.

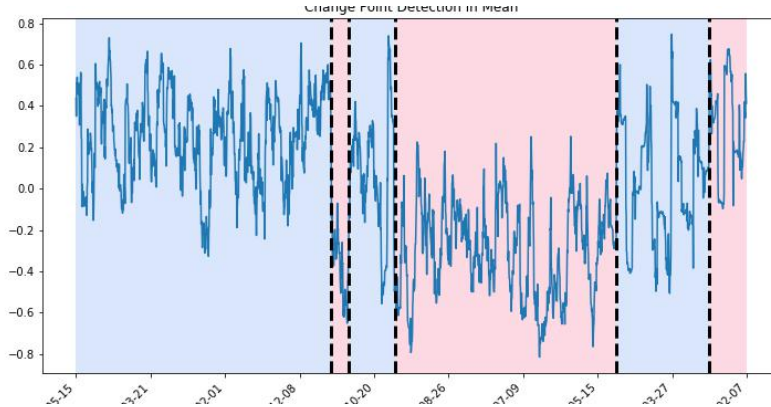
Existing application: equity-bond correlation



Existing application: Bai-Perron break dates



Existing application: average correlation by break segment



How to read a break-test result

A break test should not be used mechanically. A good empirical interpretation checks four dimensions:

1. Statistical evidence: is the test significant after allowing for unknown dates?
2. Economic narrative: does the date correspond to a plausible change in the market structure?
3. Robustness: are dates stable across trimming, frequency and specifications?
4. Decision relevance: do forecasts or portfolio weights change enough to matter?

In this class, the decision relevance is evaluated through a Markowitz portfolio.

Markov-switching models

From breaks to returning regimes

Structural breaks assume that the model changes at a small number of dates.

Markov-switching models assume that the model can revisit states:

$$y_t \mid s_t = j \sim f(y_t; \theta_j), \quad j = 1, \dots, K. \quad (28)$$

The state variable is latent:

$$s_t \in \{1, \dots, K\}. \quad (29)$$

- The data reveal probabilities of regimes, not directly observed regimes.
- This is powerful for finance: bear markets, high volatility, recession risk and correlation regimes can return.

Reference: Hamilton (1989), *Econometrica*.

The state follows a first-order Markov chain:

$$\mathbb{P}(s_t = j \mid s_{t-1} = i, s_{t-2}, \dots) = \mathbb{P}(s_t = j \mid s_{t-1} = i) = p_{ij}. \quad (30)$$

The transition matrix is:

$$P = \begin{pmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{pmatrix}, \quad p_{ij} \geq 0, \quad \sum_{j=1}^K p_{ij} = 1. \quad (31)$$

For two regimes, expected duration is:

$$D_i = \frac{1}{1 - p_{ii}}. \quad (32)$$

A simple two-state model

A useful starting point is a switching-mean and switching-variance model:

$$y_t = \mu_{s_t} + \sigma_{s_t} \varepsilon_t, \quad (33)$$

$$\varepsilon_t \sim \mathcal{N}(0, 1), \quad (34)$$

$$s_t \in \{1, 2\}. \quad (35)$$

Regime 1 can be a low-volatility expansion state and regime 2 a high-volatility recession or crisis state. A richer model adds autoregression:

$$y_t = \mu_{s_t} + \phi_{s_t} y_{t-1} + \sigma_{s_t} \varepsilon_t. \quad (36)$$

Let $\xi_{t|t}$ be the vector of filtered probabilities:

$$\xi_{t|t}(j) = \mathbb{P}(s_t = j \mid \mathcal{F}_t). \quad (37)$$

Before observing y_{t+1} , predict next-period state probabilities:

$$\xi_{t+1|t} = P' \xi_{t|t}. \quad (38)$$

This is the Markov chain part of the algorithm: today's state probabilities are propagated through the transition matrix.

Once y_{t+1} is observed, use Bayes' rule. Let η_{t+1} collect state-conditional densities:

$$\eta_{t+1}(j) = f(y_{t+1} \mid s_{t+1} = j, \mathcal{F}_t; \theta_j). \quad (39)$$

The filtered probabilities are updated as:

$$\xi_{t+1|t+1} = \frac{\eta_{t+1} \odot \xi_{t+1|t}}{\mathbb{1}'(\eta_{t+1} \odot \xi_{t+1|t})}. \quad (40)$$

The denominator is the predictive density:

$$f(y_{t+1} \mid \mathcal{F}_t) = \mathbb{1}'(\eta_{t+1} \odot \xi_{t+1|t}). \quad (41)$$

The log-likelihood is the sum of the predictive log densities:

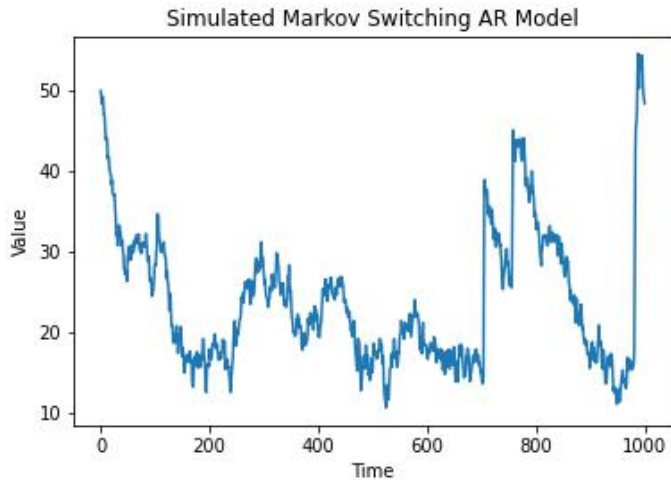
$$\ell(\theta) = \sum_{t=1}^T \log f(y_t \mid \mathcal{F}_{t-1}; \theta). \quad (42)$$

Substituting the filter:

$$\ell(\theta) = \sum_{t=1}^T \log [\mathbb{1}'(\eta_t \odot \xi_{t|t-1})]. \quad (43)$$

- Estimate θ by maximum likelihood.
- The filter is run for each trial value of θ .
- Smoothed probabilities use the full sample to infer historical states.

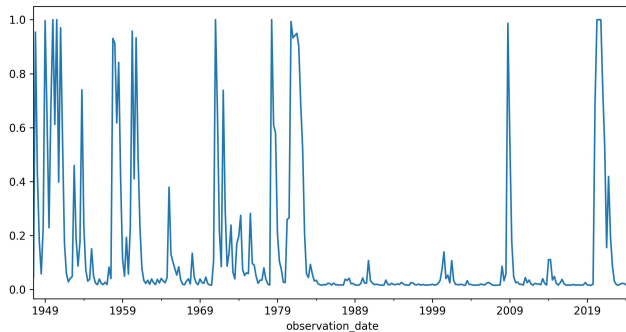
Simulation: what a Markov-switching model creates



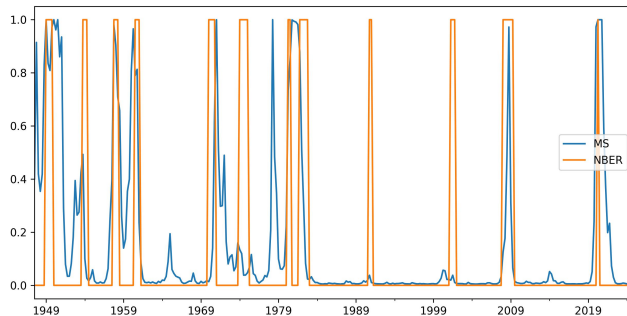
Existing application: regime switching and GDP

Regime 0 parameters						
	coef	std err	z	P> z	[0.025	0.975]
const	0.0154	0.001	17.743	0.000	0.014	0.017
sigma2	4.546e-05	6.1e-06	7.452	0.000	3.35e-05	5.74e-05
ar.L1	0.4859	0.055	8.910	0.000	0.379	0.593
Regime 1 parameters						
	coef	std err	z	P> z	[0.025	0.975]
const	0.0175	0.005	3.449	0.001	0.008	0.027
sigma2	0.0007	0.000	3.454	0.001	0.000	0.001
ar.L1	0.1031	0.162	0.638	0.524	-0.214	0.420
Regime transition parameters						
	coef	std err	z	P> z	[0.025	0.975]
p[0->0]	0.9536	0.018	52.982	0.000	0.918	0.989
p[1->0]	0.2470	0.110	2.254	0.024	0.032	0.462

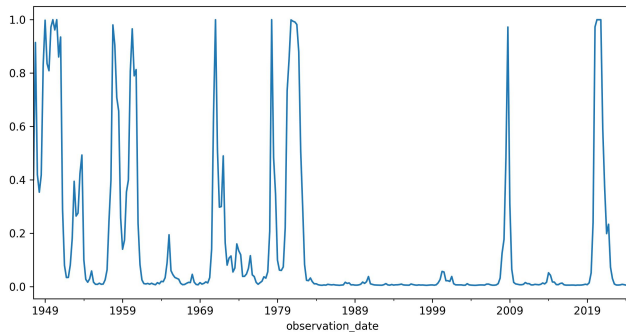
Existing application: filtered recession probability



Existing application: comparison with NBER dates



Existing application: smoothed recession probability



For returns, a simple MS model can be written as:

$$r_t = \mu_{s_t} + \sigma_{s_t} \varepsilon_t. \quad (44)$$

In many applications:

$$\mu_{bear} < \mu_{bull}, \quad \sigma_{bear} > \sigma_{bull}. \quad (45)$$

- The model transforms return observations into bear-market probabilities.
- It does not forecast crises perfectly, but it provides a probabilistic diagnostic.
- Extensions can include time-varying transition probabilities or multivariate states.

References: Maheu and McCurdy (2000); Guidolin and Timmermann (2007).

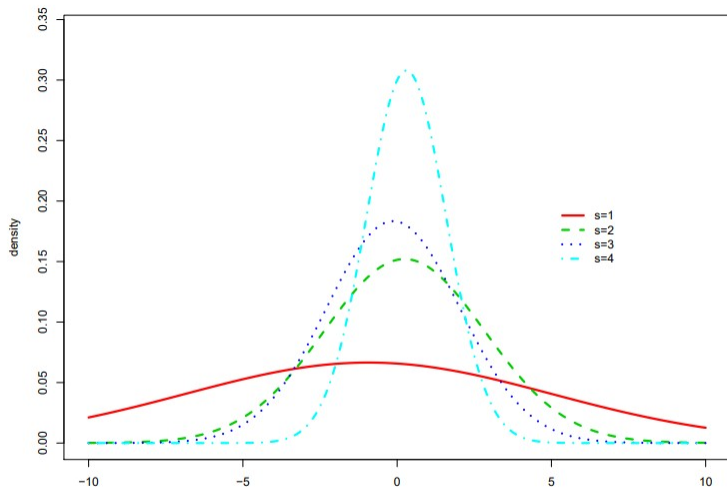
Table 4: MS-4-State Model Estimates

	mean	median	std	95% DI
μ_1	-0.94	-0.92	0.27	(-1.50, -0.45)
μ_2	0.23	0.23	0.10	(0.04, 0.43)
μ_3	-0.13	0.12	0.08	(-0.31, -0.01)
μ_4	0.30	0.29	0.04	(0.22, 0.38)
σ_1	6.01	5.98	0.35	(5.41, 6.77)
σ_2	2.63	2.61	0.18	(2.36, 3.08)
σ_3	2.18	2.19	0.12	(1.94, 2.39)
σ_4	1.30	1.30	0.04	(1.20, 1.37)
p_{11}	0.921	0.923	0.020	(0.877, 0.955)
p_{12}	0.076	0.074	0.020	(0.042, 0.120)
	0.022	0.021	0.024	(0.000, 0.048)

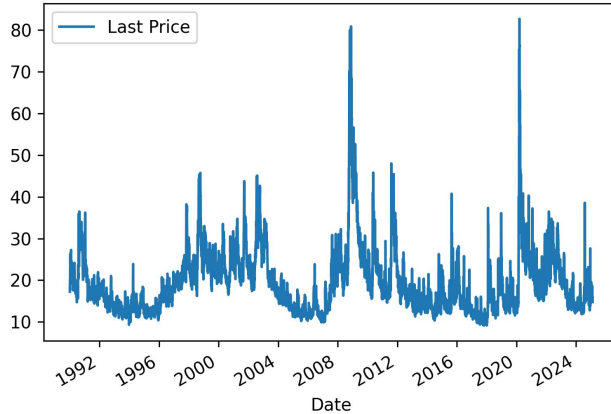
Table 8: Log Marginal Likelihoods: Alternative Models

Model	$\log f(Y \mid \text{Model})$
Constant mean with constant variance	-14924.1
Constant mean with 4-state i.i.d variance	-14256.7
MS-4-state mean with 4-state i.i.d. variance (4.10 with $K = 4$)	-14036.4
MS-2-state (4.1)	-13903.3
MS-4-state mean with constant intra-regime variance ($\sigma_1^2 = \sigma_2^2, \sigma_3^2 = \sigma_4^2$)	-13849.9
MS-4-state (4.3) with unrestricted transition matrix P	-13747.3
MS-4-state (4.3)-(4.6)	-13740.4

Existing application: market-regime probabilities



Existing application: VIX regimes



- Number of regimes: selected with information criteria, likelihood comparisons and economic interpretability.
- Label switching: state names are arbitrary; label them after estimation using means, variances or correlations.
- Local maxima: estimation can depend on starting values.
- Short samples: regimes may be hard to separate.
- Decision use: regime probabilities should improve a forecast or decision, not only fit history.

Application: correlation regimes and Markowitz portfolios

Motivation: correlation is a portfolio parameter

In a mean-variance framework, the portfolio variance is:

$$\sigma_p^2 = w' \Sigma w. \quad (46)$$

For two assets:

$$\sigma_p^2 = w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1 w_2 \rho_{12} \sigma_1 \sigma_2. \quad (47)$$

- If ρ_{12} rises, diversification falls.
- If diversification falls, optimal weights must change.
- This is exactly why structural breaks and regimes matter for asset allocation.

Markowitz problem used in the application

We solve a long-only target-return Markowitz problem in each regime:

$$\hat{w}_s = \arg \min_w w' \hat{\Sigma}_s w, \quad (48)$$

subject to:

$$\mathbf{1}' w = 1, \quad (49)$$

$$\hat{\mu}' w = \mu^*, \quad (50)$$

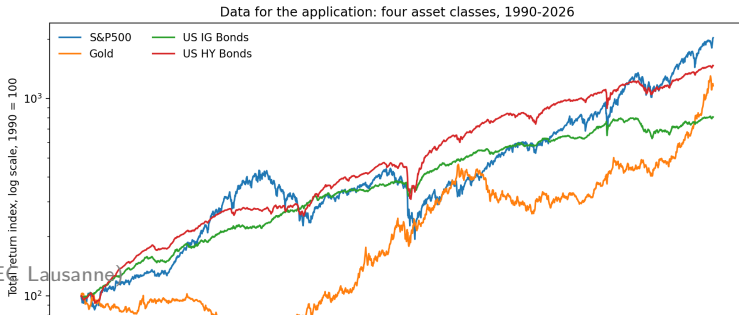
$$0 \leq w_i \leq 70\%. \quad (51)$$

- Expected returns are kept constant across regimes.
- Only the covariance matrix changes.
- This isolates the portfolio impact of volatility and correlation regimes.

Data used for the application

The application uses weekly returns computed from the file `Data_regimes.xlsx`:

- S&P 500;
- Gold;
- US investment-grade bonds;
- US high-yield bonds.



Step 1: model the equity-bond correlation regime

We use the 52-week rolling correlation between S&P 500 and US IG bonds:

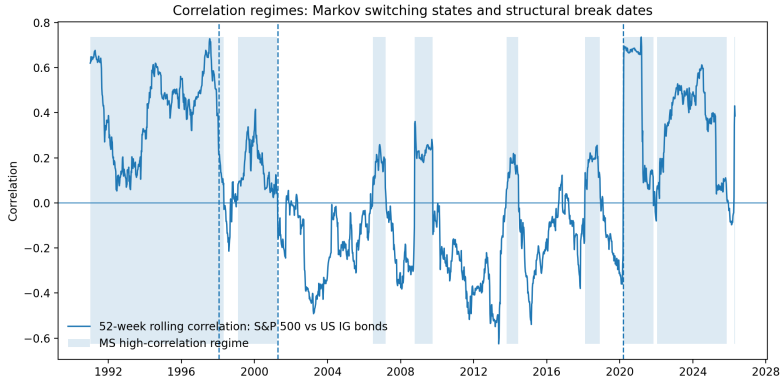
$$c_t = \hat{\rho}_{SPX,IG,t}^{52w}. \quad (52)$$

Then we estimate a two-state model:

$$c_t = \mu_{s_t} + \sigma_{s_t} \varepsilon_t, \quad s_t \in \{1, 2\}. \quad (53)$$

The same series is also used in a three-break Bai-Perron-style mean-break model.

Correlation regimes: MS states and structural breaks



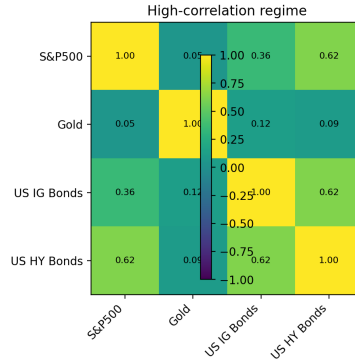
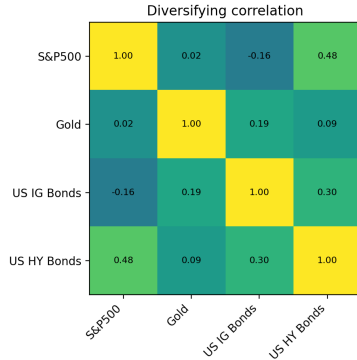
Application results: estimated regimes

Regime	Obs.	Avg. 52w corr.	SPX-IG corr.	Portfolio vol.
Diversifying correlation	891	-0.20	-0.16	4.7%
High-correlation regime	952	0.33	0.36	6.3%

- The high-correlation state is not only about correlation: it also produces a higher portfolio risk for the same target return.
- The break model isolates visible long phases: 1990-1998, 1998-2001, 2001-2020 and 2020-2026.

Correlation matrices by regime

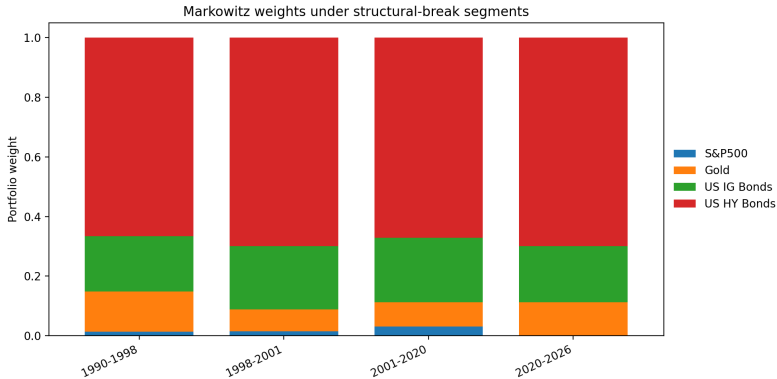
Correlation matrices change materially across regimes



Markowitz weights under Markov-switching regimes



Markowitz weights under break segments



What the application teaches

1. The covariance matrix is not stable through time.
2. Markov switching treats high and low correlations as recurring states.
3. Structural breaks treat correlation changes as persistent shifts.
4. Both approaches change the optimal Markowitz weights because $\hat{\Sigma}$ changes.
5. For portfolio construction, regime diagnostics are useful only if they change a decision.

The notebook attached to the class implements the full empirical application:

- data loading and weekly return calculation;
- rolling equity-bond correlation;
- two-state Markov-switching estimation;
- Bai-Perron-style dynamic programming for mean breaks;
- long-only Markowitz optimization by regime and by break segment.

Pedagogical point: the notebook deliberately avoids black-box portfolio code. The optimization problem is written explicitly with `scipy.optimize`.

Conclusion

- Structural breaks test whether parameters changed permanently at known or unknown dates.
- Bai-Perron extends the logic to multiple unknown breaks and estimates the dates globally.
- Markov-switching models replace fixed break dates with latent returning states.
- In finance, volatility and correlation regimes often matter more than mean regimes.
- Portfolio optimization is highly sensitive to regime changes because the covariance matrix is an input to the Markowitz problem.

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